

Production Architecture

GEX Dashboard Live Event-Option Stack

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Abstract

This document describes the current production architecture. The active trading path is the JEPA event-option static-union package for 0DTE SPXW, QQQ, and SPY options. The older Hybrid Attention-MLP, GBT+RL, level-stability, structural-profile, OptionValue, and 180 minute JEPA systems remain in the repository as research or legacy tooling, but they are not the active live contract unless a service file explicitly points to them.

1 Active Runtime

The live trading stack has two cooperating services:

Component	Role
<code>services/realtime_feed.py</code>	Downloads and normalizes live spot and option-chain data. It writes 0DTE snapshots and feature files under <code>rt_data/YYYYMMDD/</code> .
<code>bots/tradingbot_wrapper_jepa.py</code>	Loads the event-option policy and component registry, scores live candidates, manages open positions, and emits paper order intents.
<code>systemd/realtime_feed.service</code>	Production unit for the feed.
<code>systemd/ai_bot.service</code>	Production unit for the bot.

The bot is an alert and paper-intent process. It does not submit broker orders.

2 Production Package

The active model package is:

```
neural/models/jepa/jepa_production_event_options_frozen2025_static_union_202607/
  event_option_policy.json
  component_registry.json
```

The policy id is:

```
event_option_frozen2025_static_union_balanced_202607
```

The validated result directory is:

```
research_papers/JEPA/results/
  event_option_mh30trail_frozen2025_static_union_deploy202607_production_v1
```

3 Data Flow

ThetaData / live market data → realtime feed → event-option snapshot
→ live scorer → paper order intent → Discord / tracker

The feed writes files such as:

```
rt_data/YYYYMMDD/event_option_snapshots_latest.parquet
rt_data/YYYYMMDD/spot_SPX_latest.parquet
rt_data/YYYYMMDD/SPXW_greeks_0dte_latest.parquet
rt_data/YYYYMMDD/SPXW_ohlc_0dte_latest.parquet
```

The bot writes files such as:

```
trades_jepa/open_positions_jepa.json
trades_jepa/event_option_runtime_state.json
trades_jepa/event_option_candidate_audit_jepa.jsonl
trades_jepa/paper_order_intents_jepa.jsonl
trades_jepa/tradingbot_jepa.log
```

4 Execution Contract

Field	Current value
Tickers	SPX, QQQ, SPY
Option symbols	SPX maps to SPXW; QQQ and SPY trade their own option chains.
Expiry	0DTE only
Entry window	10:00–14:30 ET
EOD cleanup	16:00 ET
Risk capital	\$5,000 per trade
Stop	-60%
Trail	Activate at +50%, close after 25% giveback from high water mark
Emergency take profit	+1000%
Hold bounds	Minimum 30 minutes, maximum 180 minutes
Broker mode	Paper intents only

5 Static Policy Caps

Ticker	Bucket	Min score	Max trades/day	Cooldown
SPXW	d25	0.34	4	0m
QQQ	d35	policy	2	30m
SPY	d35	policy	1	0m

6 Validation Snapshot

Completed-month validation over January through June 2026:

Scope	Trades	WR	PF	Min month trades
Overall	697	57.819%	1.915	103
QQQ	236	60.593%	1.771	36
SPXW	355	55.211%	2.037	51
SPY	106	60.377%	1.769	12

The deployable gate uses a minimum of 12 trades per completed month. The stronger preferred target of 18 trades per month per ticker is not met by SPY in this package.

7 Legacy Systems

The following systems are retained for research and comparison:

- GBT and PPO/RL in `neural/` and `neural/rl/`;
- Hybrid Attention-MLP artifacts;

- 180 minute JEPa direction models;
- level-stability and structural-profile policies;
- OptionValueJEPa strike/exit research.

They should not be treated as production unless a new promotion changes the systemd units and passes the live-ready validation contract.